Detection of Anomalous Behavior In An Examination Hall Towards Automated Proctoring

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Abstract— **The paper proposes a workflow for the automatic detection of anomalous behavior in an examination hall, towards the automated proctoring of tests in classes. Certain assumptions about normal behavior in the context of proctoring exams are made. Anomalies are behavior patterns that are relatively (and significantly) different. While not every anomalous behavior may be cause for suspicion, the system is designed to detection typical patterns for actions of concern such as discussions during an exam or the turning around or the passing of notes, etc. This detection is based on features computed using the histogram of gradient orientations followed by a nearest -neighbor search through annotated patterns of pre- recorded clips to train the system for behavior that may cause concern. While there may be false positives, the system is intended as a decision support system to facilitate automatic proctoring of tests and deters malpractice.**

Keywords—: Hog features; KNN classifier; Spatio-temporal context; Video Processing; Behavior recognition; Anomaly detection

I. Introduction

In recent years, with the rapid development of computer technology, video surveillance technology has made considerable progress in many areas. But the traditional method of video surveillance is fulfilled by personnel monitoring, which is a heavy workload.

Video Analytics is a technology that is used to analyze video for specific data, behavior, objects or attitude. This paper studies the automatic surveillance methodology in Examination hall, which can detect abnormal behavior in real time. It assumes that for any specific context, there is a notion of what constitutes normal behavior and conversely abnormal behavior. Anomalies stand out to be as different relative to the context of their surrounding in space of time. Hence it is a good way to solve this problem using this technology [1].

Video surveillance can be an effective tool for today's businesses both large and small such as in video surveillance in **examination** halls, security surveillance and deterring dishonest and deceitful behaviors [2].
For such systems it is needed to design a core which uses a method to detect human actions, classify them based upon several actions in sequence.

However, building an abnormal behavior recognition system is a challenging problem because of the variations in the quality of the video, environment size and certain postures of the humans. Environment of the examination room is crowded and dynamic, which imposes challenge for current approaches to video action detection because it is difficult to segment the student from the background due to disrupting motion from other objects and the scene. All of these make it difficult to satisfy the application in the real-world scene [5].

Some of the main applications of Human Behavior detection are crime detection in sparsely populated areas like ATMs, automated sports commentary, intrusion detection, and detection of jaywalking on roads, characterization of human gait, person counting in a crowd, gender classification and fall detection for elderly people. In today's world with increase in technology, systems can be developed and used to detect human activity. This paper aims at detecting human activities which are then classified into different categories. These categories are then studied, identified and interpreted as normal or abnormal behavior. The aim focuses on developing this system to urge an increase in the security system and services provided for the security of examination authority.

The paper proposes a system for abnormal activity detection in examination hall videos using K Nearest Neighbour (KNN) which encode scene rules and are used to smooth sequences of actions. High-level behavior recognition is achieved by computing the probability that a set of predefined KNN's explains the present action sequence.

This system is deployed in an examination hall where the student's activities and behavior are monitored on a surveillance camera. System accepts video as the input and an automatic alerting system alerts the relevant authorities when required. Based on the predefined anomalies, the activities are categorized as normal or abnormal behavior.

A.Literature Survey

Most of the research work in the field of video surveillance systems has become very popular due to high security concerns and low-hardware costs which have been carried out in the last decade. This section presents the ones which are more relevant to this work. Many issues such as behavior understanding in crowded scenes, nurture care institutions, law application, building security and traffic analysis.

Surveillance videos have shown to capture the actions of the subjects that either display normal or anomalous behavior. Object recognition, specifically face recognition has been implemented using the Viola Jones algorithm. The two activities that were focused on were passing cheats and viewing neighbor's answer sheet. Interest points are measured from the individual frames using SURF and interest points are detected using the Hessian Matrix Approximation. The actions of human behavior are viewed as a stochastic order of actions [4].

In the environment of examination halls, tracking of behavior and training the system for various anomalies is critical to avoid any false negatives. Behavior detection has been implemented using spatial-temporal shape and flow correlation [5]. The training templates are extracted and used as a reference. Mean shift algorithm is implemented to segment the video. Each frame of the input video is compared with the template image and the matching distance is calculated. Setting an appropriate threshold ensures location of potential matches in the video.

Another condition in video-surveillance is in the outdoor environments. It focuses on detecting mobile objects in outdoor scenes. A reference image is used that contains the features that are well trackable. A feature correspondence is found between pairs of adjacent frames from the input video. Motion detection is tracked using the Lucas-Kanade tracker [6]. Bad tracked features are rejected and new features that contribute to new image areas are included, for better performance. Thus, this system depends on the selection of a set of features that are robust and traced to anomalous behavior; this is a challenge as the specific patterns may well depend on the conditions under which the video is recorded.

In most of the contexts it is necessary for us to analyze the videos in terms of detailed features, which is inefficient in terms of accuracy and cost. A lot of research work is going on in the field of detection of abnormal behavior in examination halls. In examination halls, abnormal behavior can be detected by specifying the abnormal/suspicious activities and then detecting them. The anomalies are then identified and reported to the necessary authorities. The paper has automated the system to detect any human behavior remotely in the Examination Hall. Once the systems are given the video feed it detects the actions automatically. The system can then immediately alert the relevant authorities who can respond to the action appropriately [7].

II. PROPOSED METHOD

The system is made to be adaptable to different conditions, meaning the user can choose the behaviors they want the system to detect and train the system specifically for that. This means that it can be used across a variety of situations and conditions from normal surveillance to security related monitoring.

This paper presents a method for anomaly detection in videos based on HOG (Histogram of Oriented Gradients) features [8].

A. Design Considerations

This section deals with mentioning various assumptions taken into consideration for the system to work as desired. The various dependencies such as minimum resolution of the input video, video format and other parameters are discussed here. Pruning techniques are used to overcome the challenges like background clutter, variation in object scale etc.

B. Assumptions and Dependencies

Some of the assumptions made in designing the system are: the camera position is fixed and stationary, the video provided preferably be in gray scale and the input video should not have a very high resolution > 500x500.

C. Goals and Constraints

- The primary goal is to detect suspicious human behavior from surveillance videos.
- The suspicious behavior is then flagged and reported.
- Suspicious actions that are not trained prior cannot be detected.
- The system wouldn't work well in a crowded environment.

The systems was studied and implemented the automatic surveillance methodology in Examination hall, which can detect abnormal behavior in real time. It is assumed that for any specific context, there is a notion of what constitutes normal behavior and conversely abnormal behavior. Anomalies stand out to be as different relative to the context of their surrounding in space of time. Figure 1: presents a schematic block diagram of the proposed system. The system contains three main modules – the Detection, Recognition and Classification modules.

Fig 1: A schematic block diagram of the proposed system

1. Detection

The input module accepts the real time videos input from the users. In the subsequent steps, operations to classify the different actions like peeking, turning around, passing paper and signaling in the video are performed.

The first step in the process is to detect all the humans that are present in the video. This is done by taking the first frame in the video and passing it to a histogram of oriented gradients (HOG) descriptor. The descriptor comes in with a default people detector also [9].

The system counts occurrences of gradient orientation in confined portions of an image. It is computed on a dense grid of consistently spaced cells and uses overlapping local contrast normalization for better accuracy.

The image is divided into small connected regions called cells, and for the pixels within each cell, a histogram of gradient directions is compiled. The descriptor is then the concatenation of these histograms.

The HOG descriptor has a few key advantages over other descriptors. Since it operates on local cells, it is invariant to geometric and photometric transformations, except for object orientation. Moreover, coarse spatial sampling, fine orientation sampling, and strong local photometric normalization permits the individual body movement of pedestrians to be ignored so long as they maintain a roughly upright position. The HOG descriptor is thus particularly suited for human detection in images.

Implementation of the HOG descriptor algorithm is as follows:

1) Divide the image into small connected regions called cells, and for each cell compute a histogram of gradient directions or edge orientations for the pixels within the cell.
2) Discretize each cell into angular bins according to the gradient orientation.
3) Each cell's pixel contributes weighted gradient to its corresponding angular bin.
4) Groups of adjacent cells are considered as spatial regions called blocks. The grouping of cells into a block is the basis for grouping and normalization of histograms.
5) Normalized group of histograms represents the block histogram. The set of these block histograms represents the descriptor.

Fig. 2 represents the implementation and features extracted from the HoG on a sample video frame.

2. Recognition

In this module, it recognizes the actions that each human performs in the video and then classify the action into either suspicious or non-suspicious behavior. KNN was used as the classifier [10].

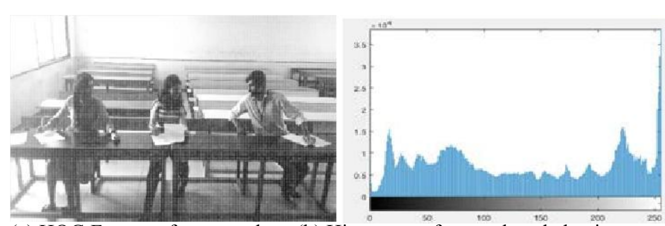
2.1 Functionality of the system

The user must be able to train the system to detect any behavior that they want to check for.

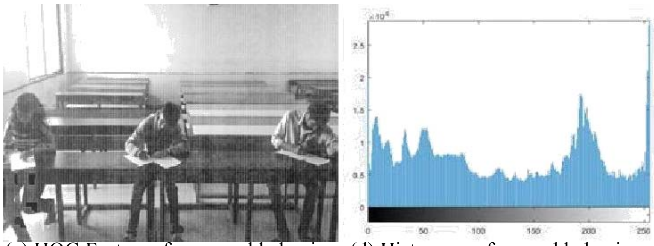
The system must be able to detect the behavior the user requires that is whether the behavior is normal or abnormal. In this system four types of abnormal behavior are recognized and they are peeking into neighbors answer sheet, passing papers, signaling to neighbors and turning around [11].

The system must inform the user when the trained behavior occurs. This is done by a message popping up on the video informing the user the name of the abnormal behavior which has occurred along with a beep sound.

The training data of this exercise is formed by a set of labeled 2D-points that belong to one of five different classes: normal, peeking, passing notes, etc., communicating through gestures (signing) and turning around.



(a) HOG Features for anomalous (b) Histogram of anomalous behavior
behavior



(c) HOG Features for normal behavior (d) Histogram of normal behavior

Fig. 2: a) HOG Features for abnormal behavior, b) Histogram of abnormal behavior, c) HOG Features for normal behavior and d) A histogram of a frame part of a video clip for normal behavior.

3. *Classification Module*

KNN classifier algorithm is used for classification . [10].

The HoG feature vector of the test image was found and the Euclidian distance to all the trained images feature vectors was calculated.

The 5 closest distances and the corresponding labels associated with it was stored and then determined which class is having a majority based on the number of occurrences of each class.

In Fig. 3,an examination hall video considering all the assumptions and dependencies was obtained. To process the video, it is broken down into frames that elicit each behavior.

Low level feature points are extracted from these frames. These feature points are then passed into a histogram of oriented gradients (HOG) descriptor.

The histogram of oriented gradients (HOG) is a feature descriptor used for the purpose of object detection. These HOG features are then given as input to the K -Nearest Neighbor (KNN) classifier which classifies based on the Euclidian distance calculated. The obtained classified frames are smoothened and finally classified actions are shown as output with a text message and a sound alert to notify the authorities [12 13].

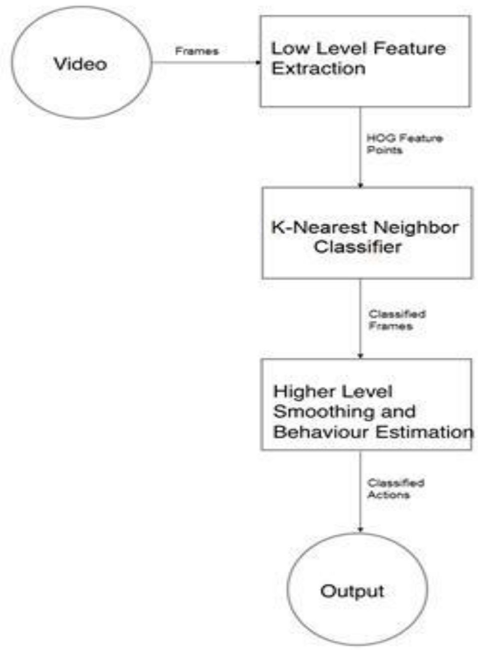


Fig .3: Block diagram representation of Classification Module.

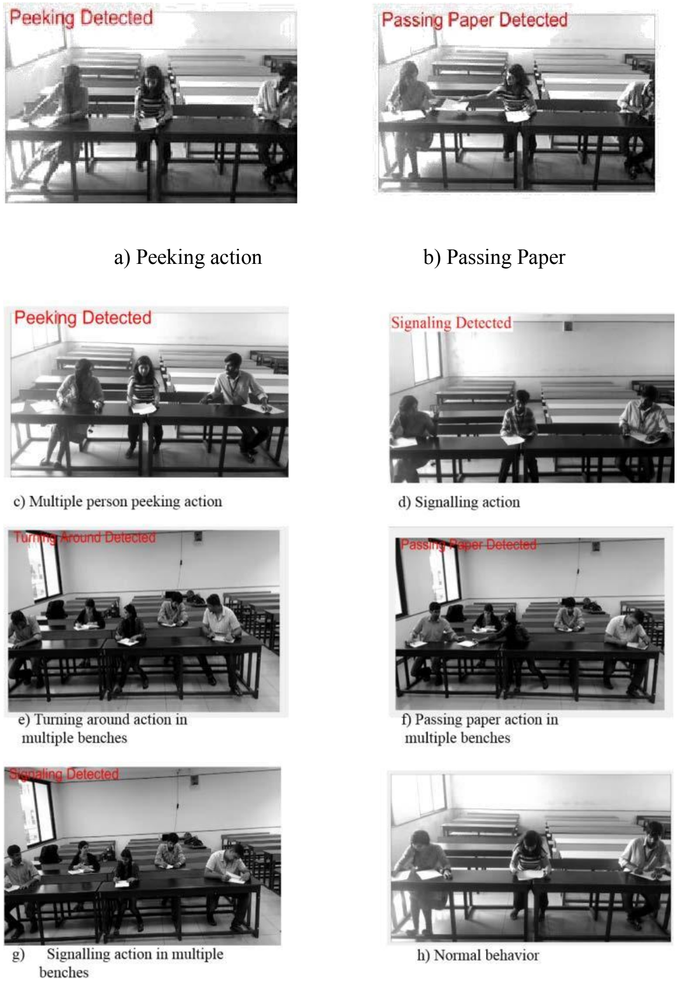
III. EXPERIMENTAL RESULTS

This paper has detected abnormal behaviors in exam hall for students sitting in a single row and also multiple rows.

A. Data Set

Training and testing videos were recorded in real time in a class room. Initially the system was tested for 3 people sitting in a single row for all abnormalities in class room such as peeking, signaling, passing paper and turn around. Later, it was also tested for students sitting in 2 rows in a class room. There videos were of resolution 720X1280 and used for training the classifier and testing different abnormalities. Using the training data it was able to distinguish normal behavior from abnormal behavior. Each video is divided in to 100 frames per second. Time taken to detect normal and abnormal activities is explained in TABLE I.

Fig. 4 (a) – (d) represent peeking, passing paper multiple peeking and signaling actions, respectively, for students sitting in a single row, Fig. 4 (e)-(g) represent turning around, passing paper and signaling actions for multiple rows of students.



a) Peeking action b) Passing Paper

c) Multiple person peeking action d) Signalling action

e) Turning around action in multiple benches f) Passing paper action in multiple benches

g) Signalling action in multiple benches h) Normal behavior

Fig.4: a) Detection of peeking action b) Detection of Passing Paper c) Detection of Multiple person Peeking action d) Detection of Signaling action e) Detection of Turning around action in Multiple benches f) Detection of passing paper action in multiple benches g) Detection of Signaling action in multiple benches H) Normal behavior

TABLE I: Time taken to detect the activities

Normal/Abnormal activity		Time taken to detect the activity
Normal		0.000003 seconds.
Abnormal	1) Peeking	1.202531 seconds.
	2) Signaling	0.827096 seconds.
	3)Turning Around	0.920118 seconds.
	4)Passing Paper	1.253411 seconds.

IV. PERFORMANCE ANALYSIS

In the area of machine learning confusion allows visualization of the performance of an algorithm. TABLE II indicates Confusion matrix of the system. Each column of the matrix represents the instances in an expected class while each row represents the instance of the actual class. Confusion matrix reports the number of false positives, false negatives, true positives and true negatives. This allows more exhaustive analysis than mere percentage of correct guesses. All the diagonal values indicate the accuracy of each action detected.

A. Evaluation Factors

In pattern recognition and information retrieval with binary classification, the performance of a model is determined by correct predictions that reflect the actual events. When a model is trained, it needs a “labeled” data set which contains the actual values which it is trying to predict. Later the model can be evaluated against labeled data which was not used to train the model.

True positives and true negatives are the observations which were correctly predicted, and therefore The terms “false positive” and “false negative” can be puzzling. False negatives are observations where the actual event was positive. The terms refer to the interpretations and not the actual events [14]. On this basis, the Precision and Recall are defined as

Precision looks at the ratio of correct positive observations. denominator is the count of all positive predictions, including positive observations of events which were in fact negative.

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \qquad (1)$$

Recall is also known as sensitivity or true positive rate. It’s the ratio of correctly predicted positive events. denominator is the count of all positive events, regardless whether they were correctly predicted by the model.

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \qquad (2)$$

Accuracy is perhaps the most intuitive performance measure. It is simply the ratio of correctly predicted observations.

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \qquad (3)$$

Equations (1) and (2) represents precision and recall formulas. Equation (3) represents accuracy formula
TABLE III represents Recall and Precision values for the actions detected. It is noted that the action ‘Passing paper’ has the highest recall and precision rates. This is explained by the fact the action is distinct. On the other hand, an action such as ‘Signaling’ could be mistaken for other actions or even hand gestures as a person speaks. Thus, there are spurious detections leading to a relatively poor precision [15 16].

TABLE II: Confusion Matrix

Predicted → / Actual ↓	Normal	Turning Around	Passing Paper	Peeping	Signaling
Normal	90.00%	0.00%	0.00%	7.00%	3.00%
Turning Around	0.00%	85.00%	1.5%	5.65%	7.85%
Passing Paper	3.00%	2.00%	94.00%	1.00%	0.00%
Peeping	5.45%	1.50%	2.5%	90.00%	0.55%
Signalling	7.00%	3.50%	1.50%	5.60%	82.40%

TABLE III: Recall and Precision values

Actions	Re-call %	Precision %
Normal	85.3%	90%
Turning Around	92.3%	85%
Passing Paper	94.4%	94%
Peeping	88.01%	90%
Signalling	87.8%	82%

The overall accuracy of the system is 88.2%.

V. CONCLUSION

This paper deals with designing an approach wherein it tries to detect any abnormal behaviors present in the videos. The system first works by detecting all students present in the video. After detecting all the students, it tracks the detected students throughout the course of the video. The features of the tracked students are calculated using HoG feature descriptor and then sent to the K-Nearest Neighbor classifier. The classifier is pre-trained to detect normal or abnormal actions. System is made to be adaptable to lots of different conditions as in, a user can choose the behaviors that they want the system to detect and train the system specifically for that. This means that it can be used across a variety of situations and conditions from normal surveillance to security related monitoring. This system is also capable of detecting abnormalities if there are more than five people in an exam hall.

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